House Price Prediction using Linear Regression

Internship Project Report:

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- *Semester: 2nd Year / 4th Semester*
- *Internship: MainCrafts Technology*
- *Submission Date: 14 July 2026*

Introduction:

This project predicts California house prices using a Linear Regression model built with scikit-learn.

Objective:

Load the dataset, perform EDA, train a regression model, evaluate performance and visualize predictions.

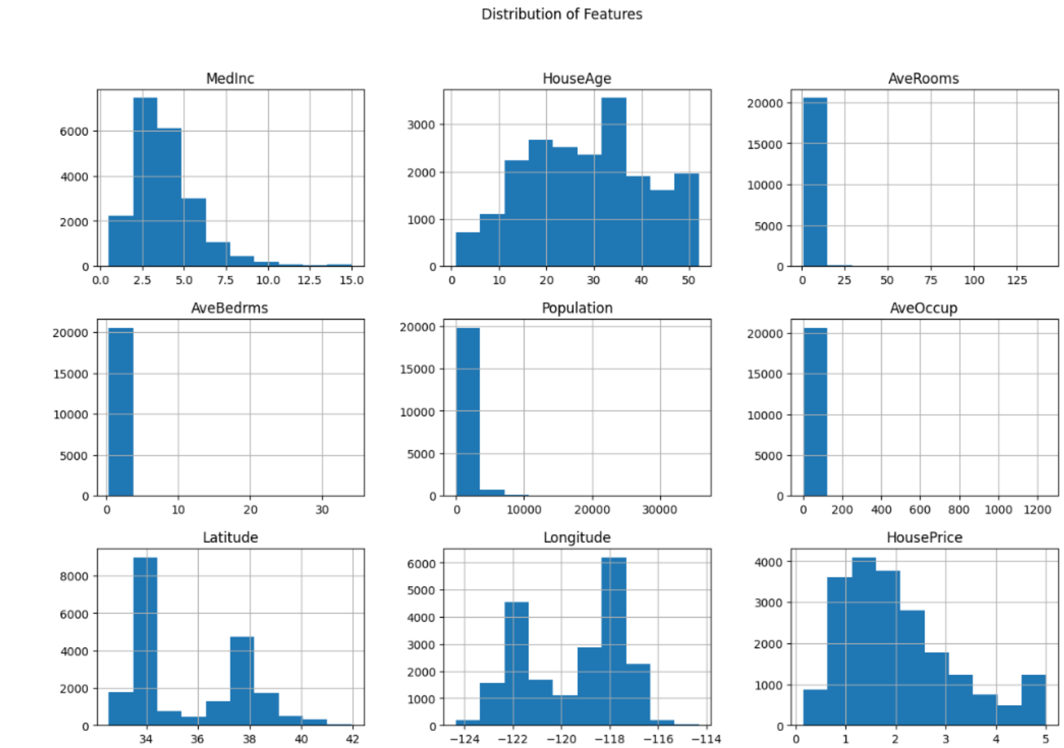
Results Model Performance:

MAE = 0.5332, RMSE = 0.7456, R^2 Score = 0.5758.

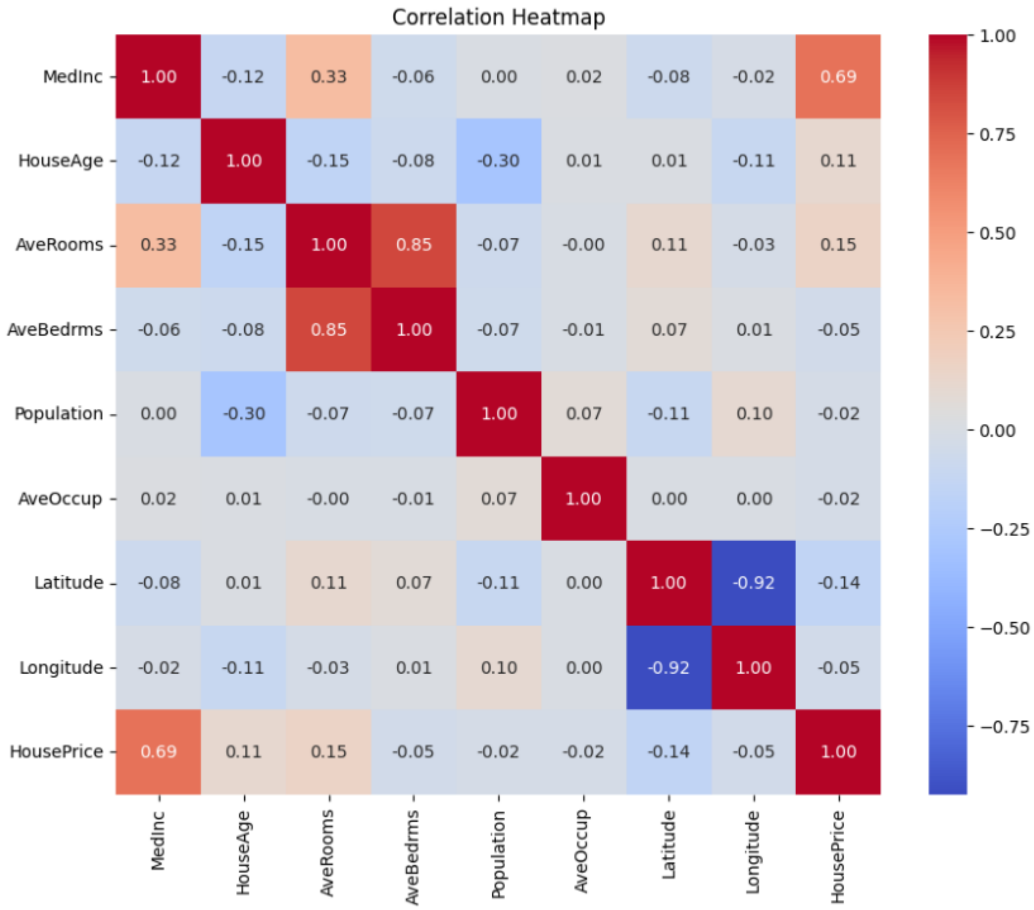
Dataset Preview:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	HousePrice
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

Feature Distribution:



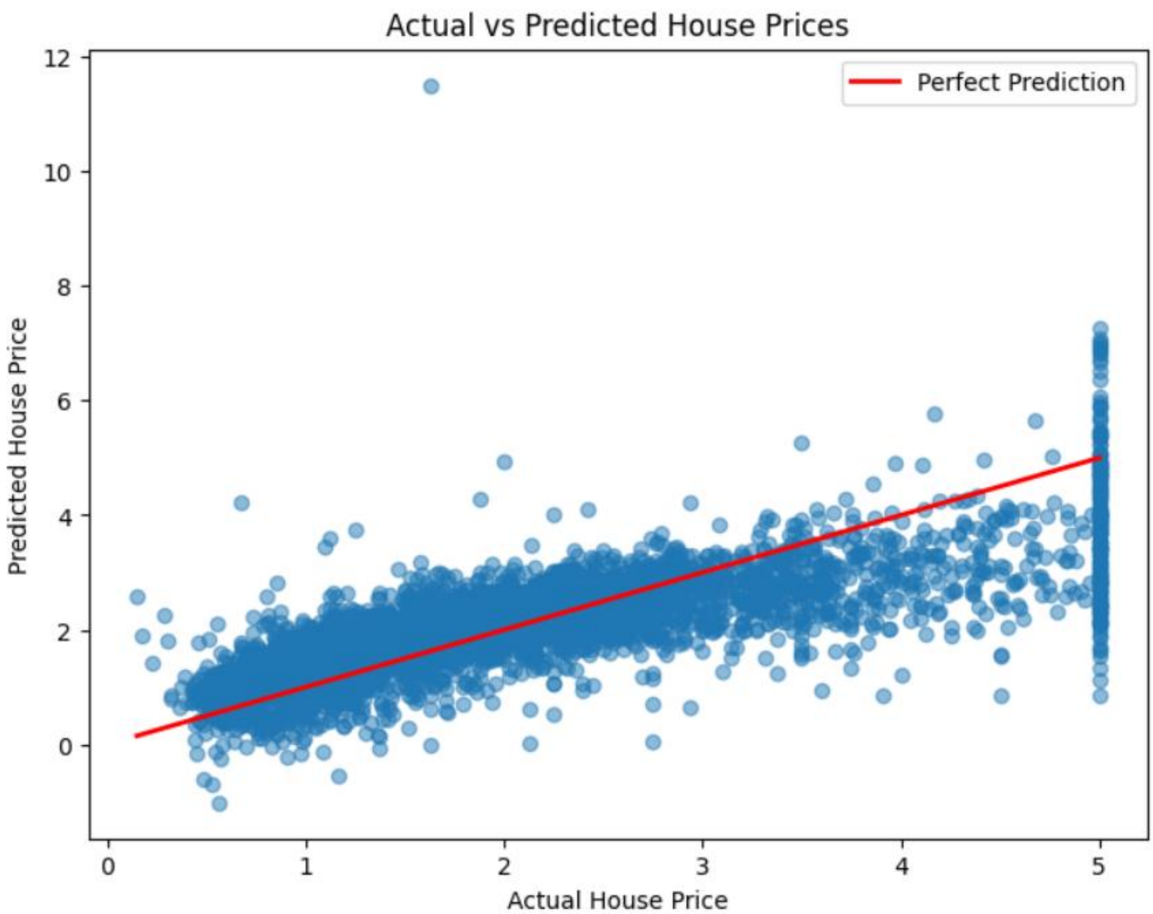
Correlation Heatmap:



Actual vs Predicted Sample:

	Actual Price	Predicted Price
0	0.47700	0.719123
1	0.45800	1.764017
2	5.00001	2.709659
3	2.18600	2.838926
4	2.78000	2.604657
5	1.58700	2.011754
6	1.98200	2.645500
7	1.57500	2.168755
8	3.40000	2.740746
9	4.46600	3.915615

Actual vs Predicted Scatter Plot:



Model Evaluation Metrics:

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MAE: 0.533200130495656
RMSE: 0.7455813830127763
R² Score: 0.5757877060324508
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Conclusion :

The Linear Regression model successfully predicts house prices and demonstrates the complete machine learning workflow including preprocessing, visualization, training, evaluation, and prediction.